FEYMAN

IM23AHRS

Product Datasheet

Inertial Measurement Unit

Document Version: V1.7

Applicable Model: IM23AHRS

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Revision History

Version	Description	Date
V1.0	Document created	2024-03-26
V1.1	Hardware information updated	2024-05-08
V1.2	English version updated	2024-06-04
V1.3	IO definition updated	2024-06-12
V1.4	IMU performance improved	2026-03-02
V1.5	IMU mechanical structure clarified	2026-03-05
V1.6	Technical specifications updated	2026-08-09
V1.7	Clarified that the AHRS roll / pitch figure is a static specification	2026-08-31

> **NOTE** This datasheet describes the IM23AHRS inertial measurement unit. All specifications are factory-calibrated across the full operating temperature range. Before production, confirm against the released firmware version and the released mechanical drawings.

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1 Product Overview

The IM23AHRS is a compact (25 × 25 × 5.9 mm) inertial measurement unit (IMU) and controller developed by FEYMAN. It is built on the mature IM19 module platform, which has been deployed across a wide range of industrial applications, with more than 100,000 units shipped by 2023. The IM23 is the mechanically reinforced version of the IM19 (surface-mount), fixed by screws, and meets the requirements of the most demanding industrial environments.

1.1 Features

- Three-axis gyroscope, three-axis accelerometer and three-axis magnetometer
- Temperature compensation and calibration
- Compact design, suited to a variety of demanding applications
- Shock resistant
- AHRS and integrated navigation algorithms (optional)

1.2 Applications

- Robotics
- Machine control
- Transportation
- Navigation

1.3 Product Appearance

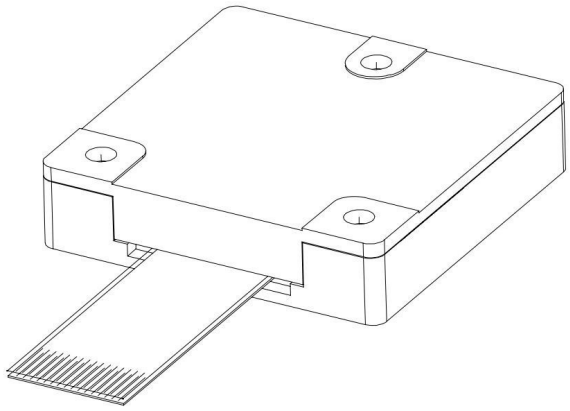


Figure 1 IM23AHRS product appearance — housing and 16-pin FPC cable

2 Specifications

GYROSCOPE	
Output range	±490 °/s
Angle random walk	0.025 °/√h
Bias instability	0.35 °/h
Scale factor nonlinearity	200 ppm
Acceleration sensitivity	1 °/h /g
Bandwidth	≤ 40 Hz, configurable

ACCELEROMETER	
Output range	±8 g
Bias repeatability (1σ)	0.5 mg
Bias instability	18 µg
Velocity random walk	0.03 m/s/√h
Scale factor nonlinearity	500 ppm
Bandwidth	≤ 40 Hz, configurable

MAGNETOMETER	
Output range	±500 µT
Resolution	0.15 µT
Temperature drift	0.027 µT/℃

Gyroscope and accelerometer bandwidth is configurable up to 40 Hz; see the IM23AHRS Software Integration Guide.

ATTITUDE	
Heading (AHRS)	0.5° (1σ)
Roll / pitch (AHRS), static	0.05°

SYSTEM	
Supply voltage	3.3 ± 0.1 V
Typical power consumption	≤ 583 mW
Dimensions	25.0 mm × 25.0 mm × 5.9 mm
Operating temperature	−40 ℃ to +85 ℃
Start-up time	< 1 s
Shock resistance	5000 g @ 0.1 ms
Vibration	20 Hz – 2000 Hz, 6.06 g
Output rate	up to 1000 Hz
FPC interface	16-pin FPC, 0.5 mm pitch

3 Hardware Definition

3.1 Pin Definition

Pin	Function	Direction	Power domain
1	**PPS-SYNC**[1]	Input	3.3 V
2	**GPIO-1**	Bidirectional	3.3 V
3	**GND**	—	GND
4	**BOOT0**	Input	3.3 V
5	**SPI-CS**	Input	3.3 V
6	**SPI-SCK**	Input	3.3 V
7	**SPI-MOSI**	Input	3.3 V
8	**SPI-MISO**	Output	3.3 V
9	**CAN-RX / GPIO2**	Bidirectional	3.3 V
10	**CAN-TX / GPIO1**	Bidirectional	3.3 V
11	**UART-RX2**	Input	3.3 V
12	**UART-TX2**	Output	3.3 V
13	**UART-RX1**	Input	3.3 V
14	**UART-TX1**	Output	3.3 V
15	**GND**	—	GND
16	**VCC**	—	3.3 V

[1] PPS-SYNC: Pin 1 connection is optional and not mandatory in AHRS applications.

3.2 Reference Circuit

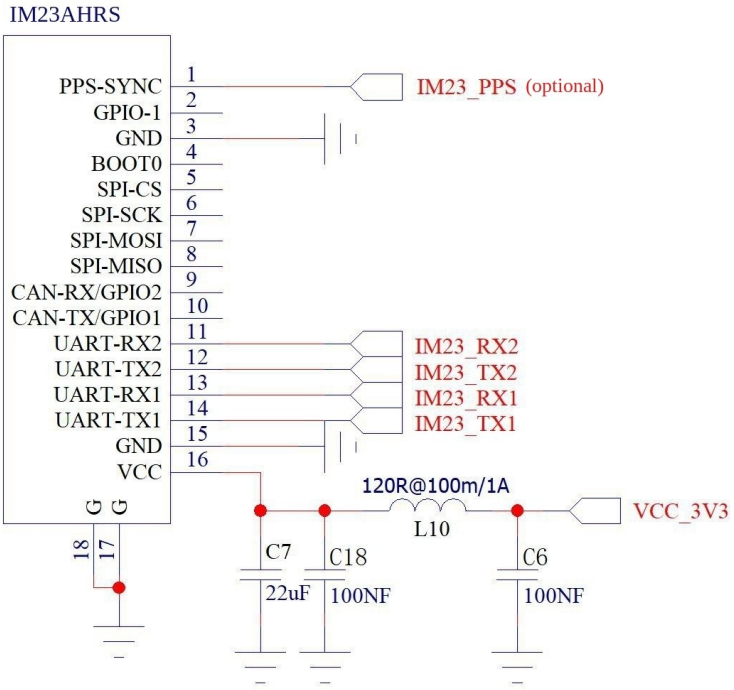


Figure 2 Reference circuit

3.3 Coordinate System

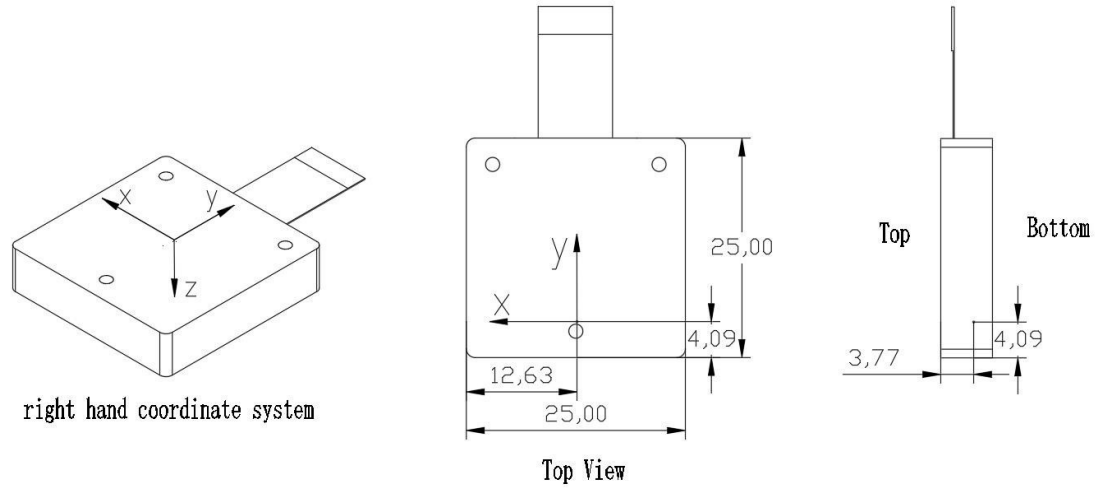


Figure 3 Coordinate system — right-hand coordinate system, top view. Dimensions in mm.

3.4 Mechanical Notes

The IM23AHRS uses a 0.5 mm pitch, 16-pin FPC interface. The reference design recommends the following mating connector: 0.5 mm pitch, insert-then-lock, dual contact on the top and bottom surfaces. Users may also select another type of FPC connector — a drawer-type connector, for example — to suit their own hardware design.

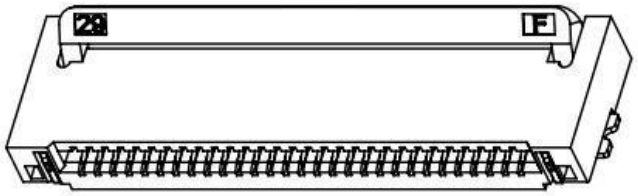


Figure 4 Recommended mating FPC connector

3.5 FPC Interface Dimensions

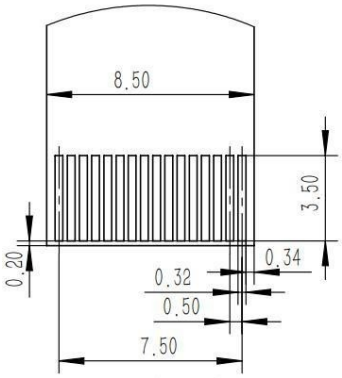


Figure 5 FPC interface dimensions. Dimensions in mm.

3.6 Outline Dimensions

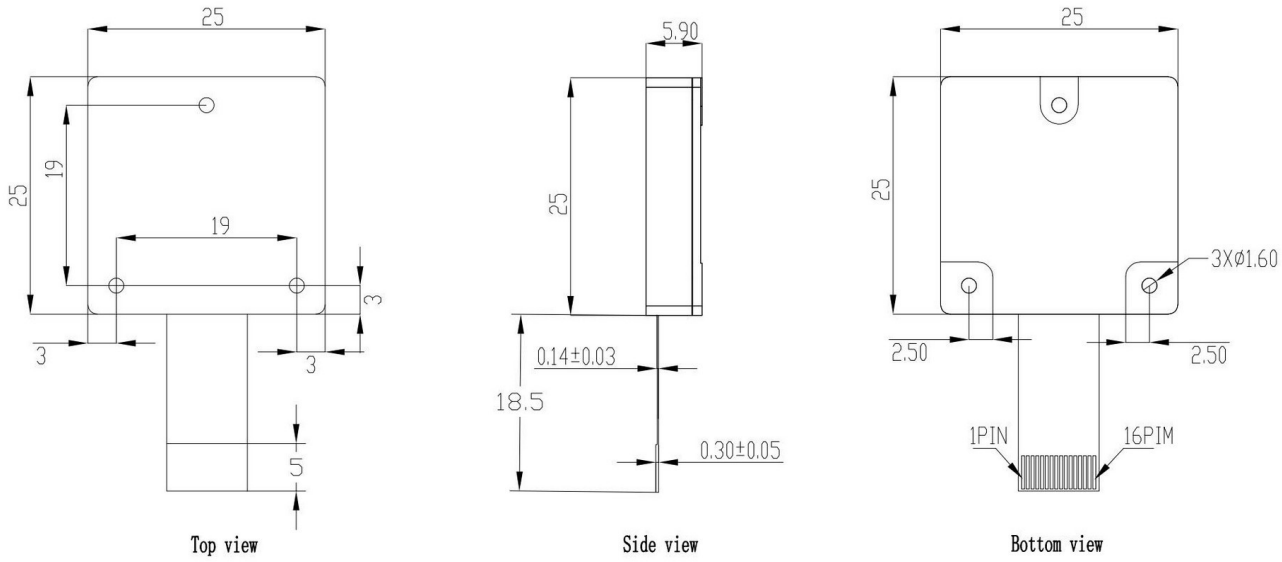


Figure 6 Outline dimensions — top view, side view, bottom view. Dimensions in mm.